Supplementary Materials for

An organic O donor for biological hydroxylation reactions

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Supplementary Materials

Figs. S1 to S3

Tables S1 to S3

Supplementary text and associated Figures (S4 to S22)

References 41-47

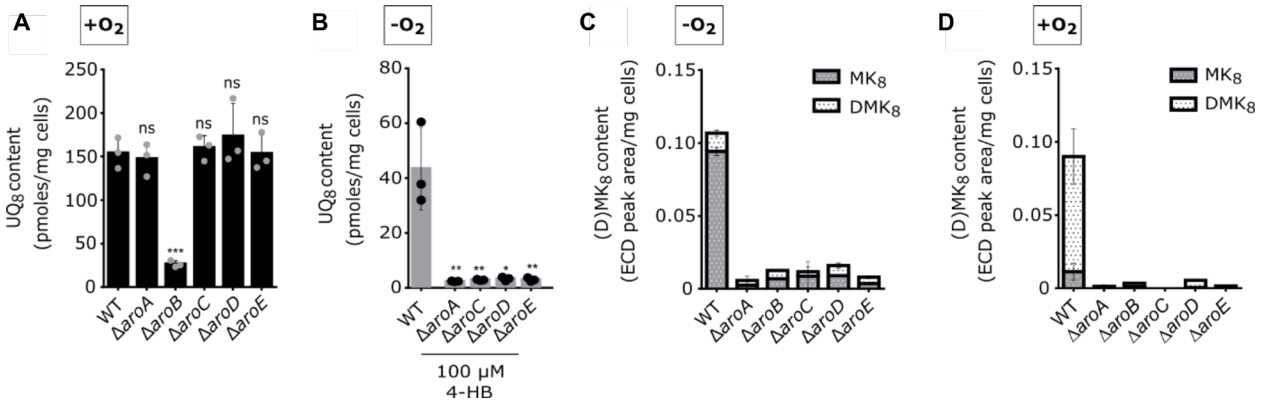


Fig. S1.

UQ_8 content in the *aro* mutant strains after aerobic growth in LB medium **(A)** and after anaerobic growth in LB medium supplemented with 100 µM 4-HB (B). MK_8 and DMK_8 content in the Δ*aro* mutant strains represented in figure 2 A-B after anaerobic growth (C) and in the Δ*aro* mutant strains represented in figure S1A after aerobic growth (D). Mean ± standard deviations (SD) (n=3). ***P < 0.001, **P < 0.01, *P < 0.05 by unpaired Student's t-test comparing to WT. ns, no significant.

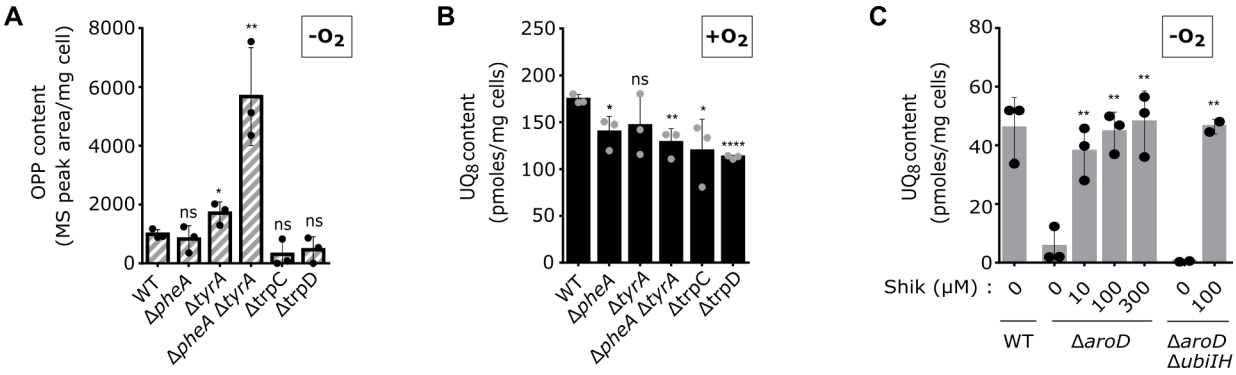


Fig. S2.

(A) OPP content in cells shown on figure 3A. UQ_8 content in the same strains than on figure 3A after aerobic growth in LB medium (B) and in Δ*aroD* and Δ*aroD* Δ*ubiIH* cells after anaerobic growth in LB medium supplemented with 10, 100 or 300 µM of shikimate (C). Mean ± SD (n=3). ****P < 0.0001, **P < 0.01, *P < 0.05 by unpaired Student's t-test comparing to WT (A-B) and to mutant strains without added shikimate (C). ns, no significant; Shik, shikimate.

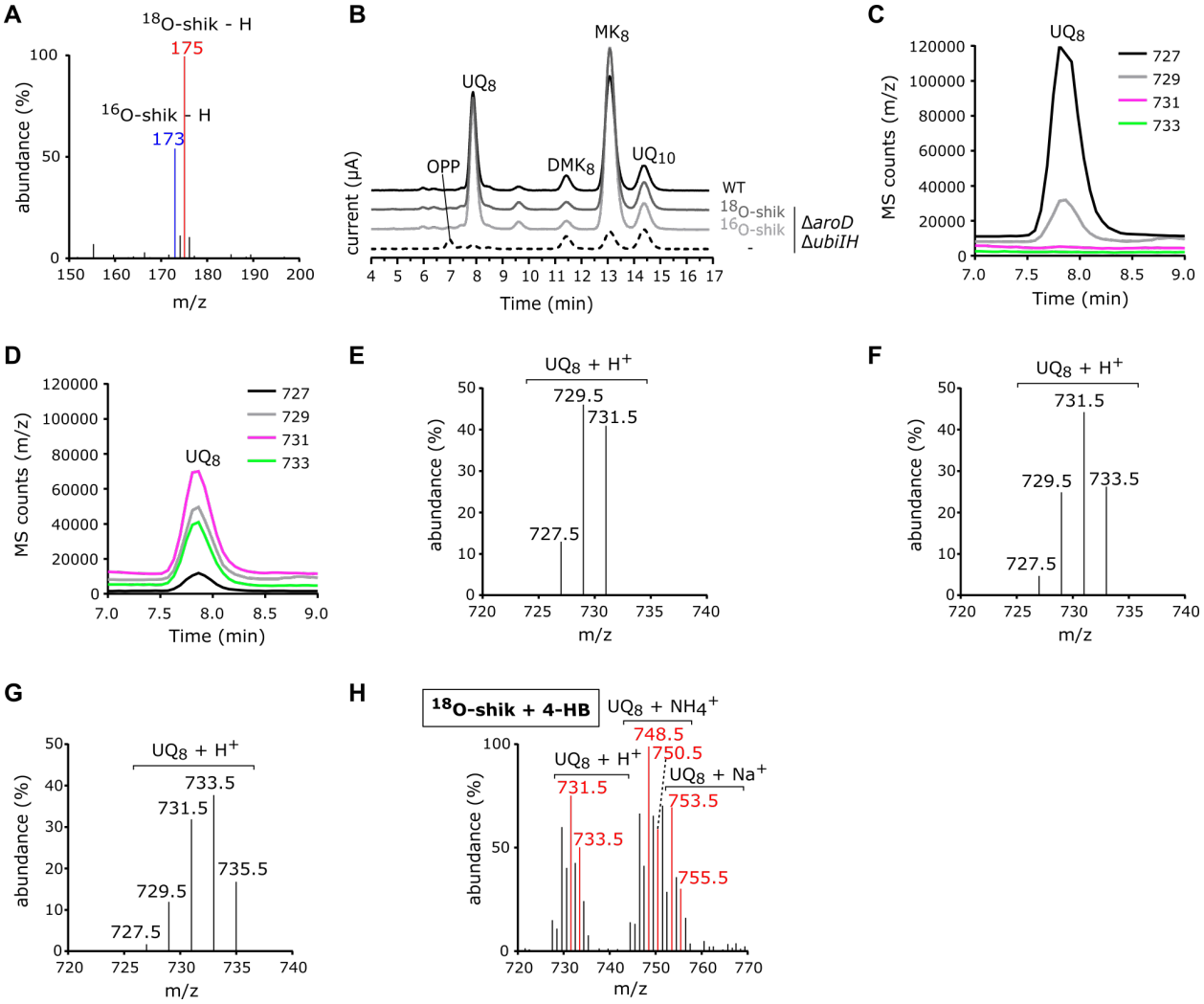


Fig. S3.

(A) Mass spectrum of synthetized C_4[^{18}O]-shikimate with the peaks corresponding to the unlabeled (^{16}O-shik) and labeled (^{18}O-shik) compounds represented in blue and red, respectively. (B) HPLC-ECD analysis of lipid extracts from 1 mg of WT and Δ*aroD* Δ*ubiIH* stains represented in figure 4B-D. The chromatograms are representative of at least three independent samples. The peaks corresponding to UQ_8, OPP, DMK_8, MK_8 and the UQ_{10} standard are indicated. (C-D) HPLC-MS profiles for indicated SIM with H^+ adduct (UQ_8 + H^+) corresponding to the analyses of figure 4B (C) and figure 4C (D). (E-G) Theoretical abundance of H^+ adducts of UQ_8 with two ^{18}O atoms($^{18}O_2$-UQ_8) (E), three ^{18}O atoms ($^{18}O_3$-UQ_8) (F), or four ^{18}O atoms ($^{18}O_4$-UQ_8) (G) based on C_4[^{18}O]-shikimate labeled at 64%. (H) Mass spectrum of UQ_8 from Δ*aroD* Δ*ubiIH* strain grown anaerobically in LB medium supplemented with 10 µM of C_4[^{18}O]-shikimate and 100 µM 4-HB corresponding to the analyses of figure 4D.

Strain	Genotype	Construction	Source
WT	MG1655		Lab strain
Δ*aroA*	BW25113Δ*aroA*		(*37*)
Δ*aroB*	BW25113Δ*aroB*		(*37*)
Δ*aroC*	BW25113Δ*aroC*		(*37*)
Δ*aroD*	BW25113Δ*aroD*		(*37*)
Δ*aroE*	BW25113Δ*aroE*		(*37*)
Δ*trpC*	BW25113Δ*trpC*		(*37*)
Δ*trpD*	BW25113Δ*trpD*		(*37*)
Δ*tyrA*	Δ*tyrA*::*kan*	Φ P1 transduction from Keio Δ*tyrA* to MG1655	This work
Δ*pheA*	Δ*pheA*::*kan*	Φ P1 transduction from Keio Δ*pheA* to MG1655	This work
FBE269	Δ*tyrA-pheA::kan*	Cassette FRT-*kan*-FRT from plasmid pKD4 recombined in MG1655 using the λRed system	This work
FBE303	Δ*tyrA-pheA*	kanamycin cassette removed from FBE269 with pCP20	This work
FBE860	Δ*ubiHI*::cat	Cassette FRT-*cat*-FRT from plasmid pKD3 recombined in MG1655 using the λRed	This work
FBE740	Δ*aroD*::*kan*	Φ P1 transduction from Keio Δ*aroD* to MG1655	This work
FBE748	Δ*ubiHI*::cat Δ*aroD*::kan	Φ P1 transduction from FBE860 in FBE740	This work
FBE843	Δ*ubiHI*::cat Δ*tyrA-pheA*	Φ P1 transduction from FBE860 in FBE303	This work
FBE1253	Δ*tyrApheA* Δ*ubiH* ::kan	Φ P1 transduction from Keio Δ*ubiH* to FBE303	This work

Table S1
Strains used in this study

Plasmid	Name	Description	Source
pCP20	pCP20	Amp, Cm, FLP recombinase expression	(*39*)
pEB268	pKD3	AmpR – FRT-cat-FRT	(*38*)
pEB269	pKD4	AmpR – FRT-kanaR-FRT	(*38*)
pEB1823	pTet	AmpR – ColE1 ori – pBAD promoter	(*15*)
pES031	pTet-*pheA*	PCR on MG1655 genomic DNA ebp28/29 (EcoRI/XhoI) in pTet (EcoRI/XhoI)	This work
pES047	pTet-*pheA*_CM	PCR on MG1655 genomic DNA ebp28/43 (EcoRI/XhoI) in pTet (EcoRI/XhoI)	This work
pES059	pTet-ubiH		(*15*)
pES077	pTet-*ubiHI*	PCR on MG1655 genomic DNA ebp61/65 (EcoRI/XhoI) in pTet (EcoRI/XhoI)	This work
pEB227	pBAD24	AmpR – ColE1 ori – pBAD promoter	(*41*)
pES154	pBAD-*ubiUV*		(*15*)
pES185	pBAD-*ubiU*(C176A)*V*		(*15*)
pEB067	pACYC184	Cam, Tet, p15a origin	(*42*)
pES232	pACYC-pBAD-*ubiUV*	pES154 (EcoRV/HindIII) in pEB067 (EcoRV/HindIII)	This work

Table S2.
Plasmids used in this study

Primers	Sequence	Use
Ebp28	ACTGAATTCATGACATCGGAAAACCCGTTAC	pheA FW
Ebp29	ACGCTCGAGTTAGGTTGGATCAACAGGCAC	pheA RV
Ebp43	ACGCTCGAGTTATGAGTGCGGATTAATTTTATTGAGATG	pheA_CM RV

Table S3.
Primers used in this study

Supplementary text and associated figures (Fig S4-S22)

Synthesis of $C_4[^{18}O]$-Shikimate

The synthesis of $C_4[^{18}O]$-shikimate was performed in five steps, starting with commercial shikimate (Figure S4). The general strategy was to protect all the reactive functions except the alcohol at position 4, in order to label it selectively with ^{18}O, and then to remove the protective groups. The first step consisted in protecting the carboxylic acid by an esterification reaction. This was carried out using methanol and *p*-toluenesulfonic acid (APTS), which acts as a catalyst. In a second step, the two alcohol functions at positions 3 and 5 were selectively protected with a silyl group. The tert-Butyldimethylsilyl group was chosen for its bulky nature and stability. The synthesis was carried out according to a slightly modified procedure previously published (*43*) by substituting tert-Butyldimethylsilyl chloride with tert-Butyldimethylsilyl trifluoromethanesulfonate (TBDMSOTf), for its high reactivity towards hindered alcohols. In a third step, the hydroxyl function at position 4 was oxidized to the corresponding ketone, using the Dess-Martin Periodinane (DMP) reagent, known for its gentle and selective conversion of alcohols into carbonyl compounds (*44*). In the fourth step, the ketone was labelled with $H_2{}^{18}O$ in an acidic aqueous medium ($CH_3C^{18}O_2H$) based on previously published work (*45*). In the fifth and final step, the selective reduction of the labelled ketone was achieved with $NaBH_4$, leading exclusively to the desired diastereomer. Finally, all the protecting groups were removed in the presence of APTS and H_2O to give the $C_4[^{18}O]$-Shikimate.

All solvents and reagents were purchased from Sigma-Aldrich and used without further purification. Dry THF was purchased under argon from Sigma-Aldrich. Reactions were monitored by thin layer chromatography (TLC) using precoated silica gel plates from Macherey-Nagel (DC-Fertigfolie POLYGRAM SIL G/UV254) followed by phosphomolybdic acid and heat as developing agents. Centrifugation was performed on Eppendorf Centrifuge 5424 R.

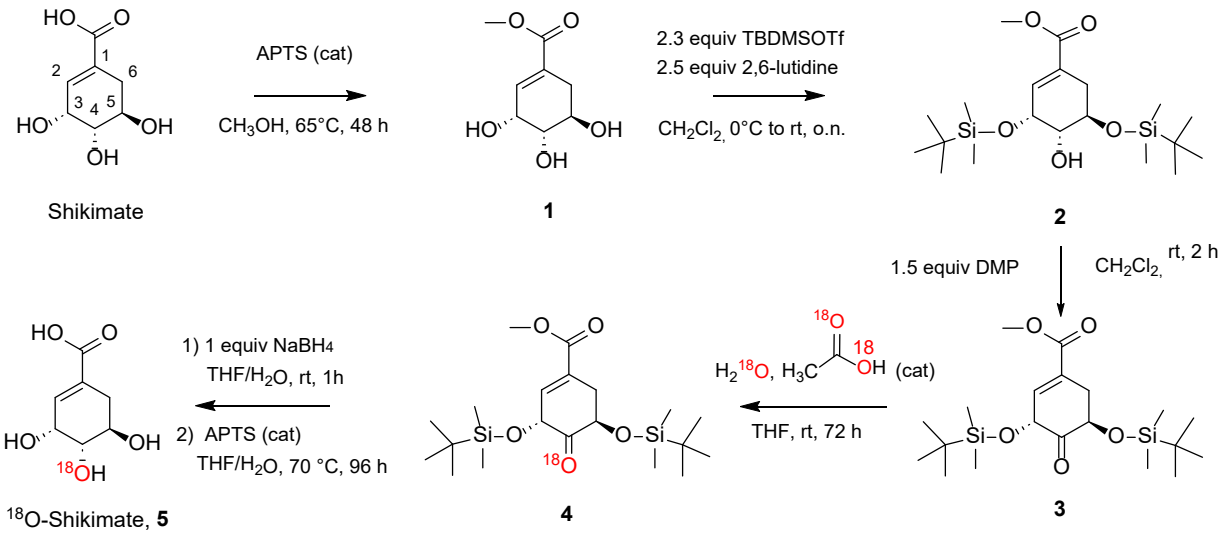


Figure S4: Synthesis of C_4[^{18}O]-Shikimate

NMR analysis: Spectra of Compounds **1**, **2**, **3**, **5** (^{18}O Shikimate) and commercial Shikimate were recorded on Bruker Advance 300 MHz NMR spectrometer. The chemicals shift (δ) are given in ppm and were measured relative to solvent residual peaks as an internal standard. For ^{1}H NMR: $CDCl_3$, δ 7.26; CD_3OD, δ 3.31. For ^{13}C NMR $CDCl_3$, δ 77.16; CD_3OD, δ 49. The data is being reported as (s = singlet, d= doublet, t = triplet, m = multiplet or unresolved, coupling constant(s) in Hz, integration, interpretation). HPLC analyses of **5** were performed on Agilent 1260 Infinity II with an Atlantis Prep T3 column (10 × 250 mm, particle size 5µM) using the following settings: flow: 2 mL/min, injection volume: 40 µL. Solvent A : H_2O/TFA 0.1% (v/v), Solvent B: Acetonitrile. An isocratic flow with 100% solvent A was performed (0-15 min) followed by a gradient (15-22 min) from 0% to 5% solvent B, and then a gradient (22-30 min) from 5 to 100 % solvent B. Elution of the HPLC column was followed at 215 nm. The final purified product was concentrated using Labconco CentriVap Concentrator connected to the Labconco CentriVap Cold Trap and the Vacuubrand Pump.

HR-MS analysis: Compounds **1**, **2**, **3**, **5** (^{18}O Shikimate) and commercial Shikimate were diluted to 1µg/mL into 1:1:4 mix of Isopropyl alcohol 20 % in water, methanol and acetonitrile before mass measurement. Each compound was nanoelectrosprayed using Nanomate (TriVersa) equipped with 5.5 µm ID nozzle chip mounted on Q exactive HF (ThermoFisher Scientific) high resolution orbitrap mass spectrometer. Masses were recorded at 60,000 resolution at 200 m/z, with an AGC set at $1^{E}6$ and with 1 µscan per scan in negative mode (except for compound **1**, recorded in positive mode).

MS/MS analysis: Commercial Shikimate and compound **5** were isolated with a 1.9 Th width and 1^E6 AGC value. Then isolated compounds were fragmented by Higher-energy collisional dissociation (HCD) with a normalized energy of collision of 10 and the resulted spectra recorded between 70 and 180 m/z.

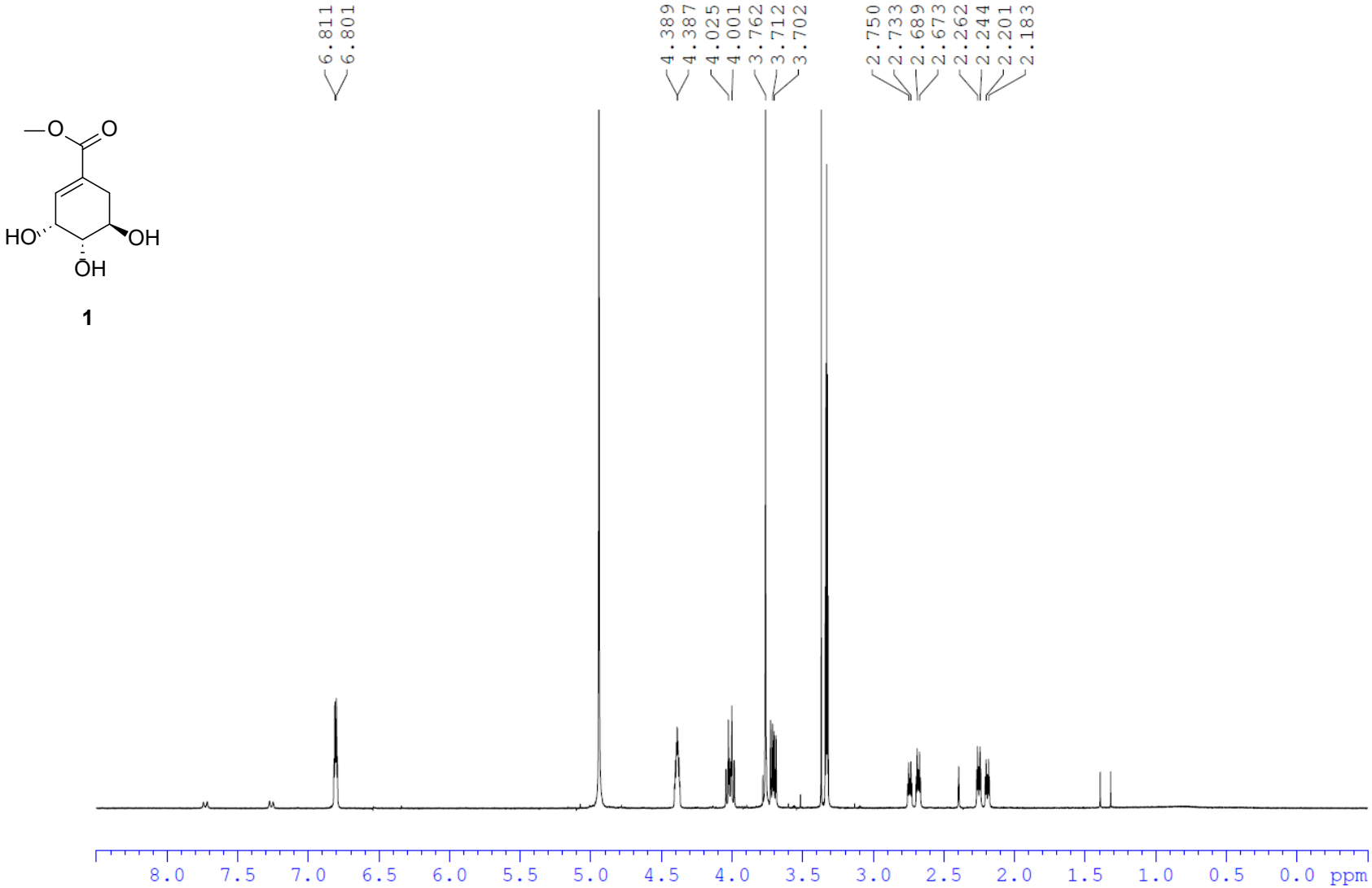
*Synthesis of methyl (3R,4S,5R)-3,4,5-trihydroxycyclohex-1-ene-1-carboxylate **(1)***:

Shikimate (1g, 5.7 mmol, 1 equiv) and methanol (25 mL) were added to a 100 mL round bottom flask equipped with a stir bar and a reflux condenser. The mixture was heated to reflux and stirred to ensure dissolution of shikimate. After complete dissolution, *p*-toluenesulfonic acid (APTS) (50mg, 0.26 mmol) was introduced as a catalyst (cat) and was heated to reflux for 2 days, after which ^{1}H NMR analysis of a reaction aliquot indicated complete conversion. Therefore, the mixture was allowed to cool to room temperature then the solvent was evaporated under reduced pressure and the residue was taken up in a minimal amount of methanol. The product was purified by pouring the reaction mixture drop by drop into excess pentane under rapid stirring and the precipitate filtered off using a Büchner vacuum filtration funnel to afford the expected product shikimate ester as a white powder (833 mg, 4.4 mmol, 77%).

Figure S5: **^{1}H NMR (300 MHz, CD_3OD, 25 °C)** δ (ppm) 6.81 (dtd, J = 3.7, 1.9, 0.7 Hz, 1H, H_2), 4.38 (m, 1H, H_3), 4.01 (dddd, J = 7.1, 5.4, 4.9, 0.6 Hz, 1H, H_5), 3.76 (s, 3H, COOMe), 3.71 (dd, J = 7.2, 4.2 Hz, 1H, H_4), 2.71 (ddtd, J = 18, 4.8, 1.8, 0.3 Hz, 1H, H_6), 2.22 (ddt, J = 18, 5.3, 1.7 Hz, 1H, $H_{6'}$).

Figure S6: **^{13}C NMR (75 MHz, CD_3OD, 25 °C)** δ (ppm) 168.14, 138.53, 129.62, 72.01, 67.83, 66.69, 51.78, 30.91.

Figure S7: **HRMS** Calculated $C_8H_{13}O_5$ $[M+H]^+$ 189.07574 | Found 189.0758



1

Figure S5: Compound **1, ^{1}H NMR (300 MHz, CD_3OD, 25 °C)**: δ (ppm) 6.81 (dtd, J = 3.7, 1.9, 0.7 Hz, 1H, H_2), 4.38 (m, 1H, H_3), 4.01 (dddd, J = 7.1, 5.4, 4.9, 0.6 Hz, 1H, H_5), 3.76 (s, 3H, COOMe), 3.71 (dd, J = 7.2, 4.2 Hz, 1H, H_4), 2.71 (ddtd, J = 18, 4.8, 1.8, 0.3 Hz, 1H, H_6), 2.22 (ddt, J = 18, 5.3, 1.7 Hz, 1H, $H_{6'}$).

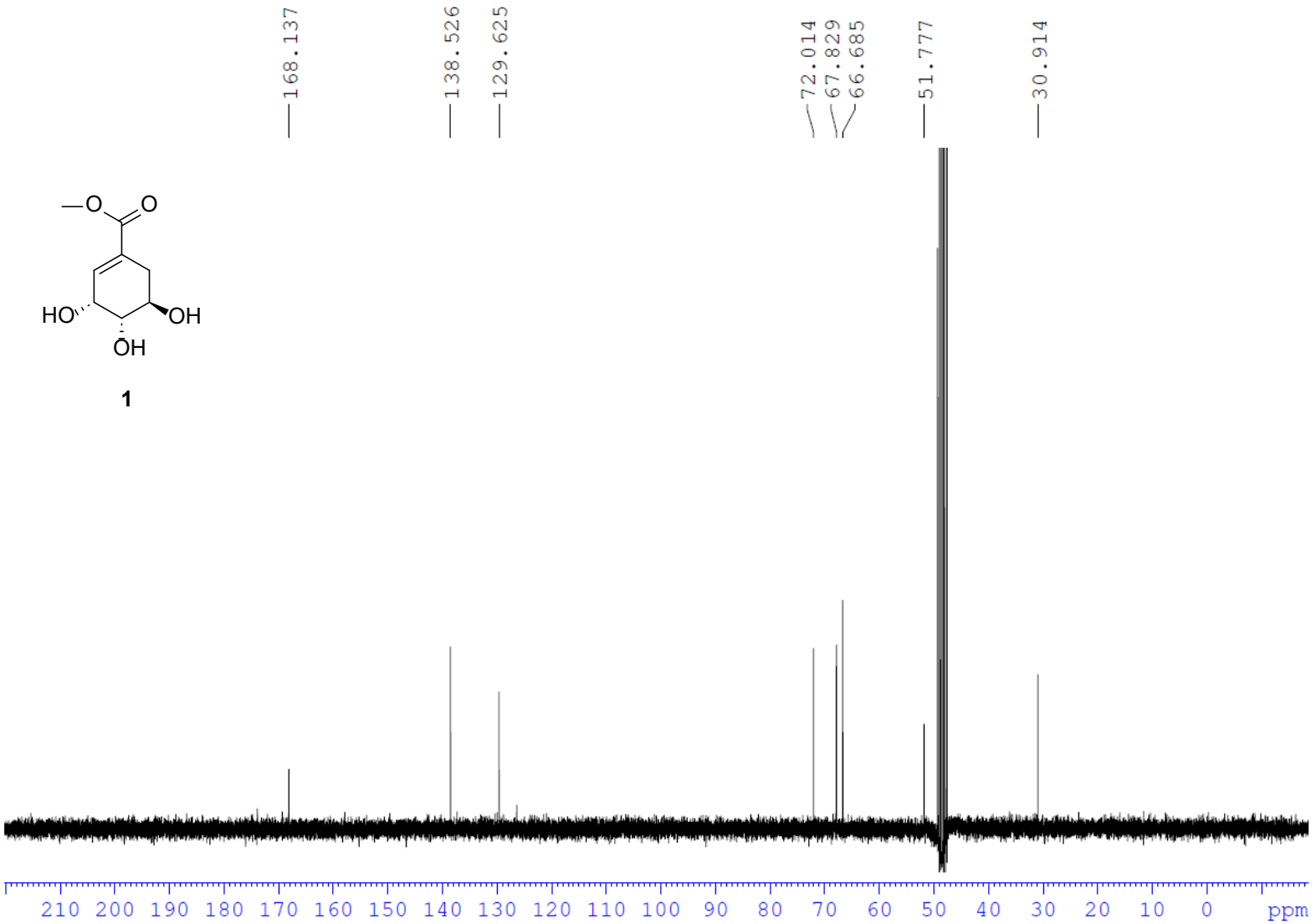


Figure S6: Compound **1, ^{13}C NMR (75 MHz, CD_3OD, 25 °C)** δ (ppm)168.14, 138.53, 129.62, 72.01, 67.83, 66.69, 51.78, 30.91.

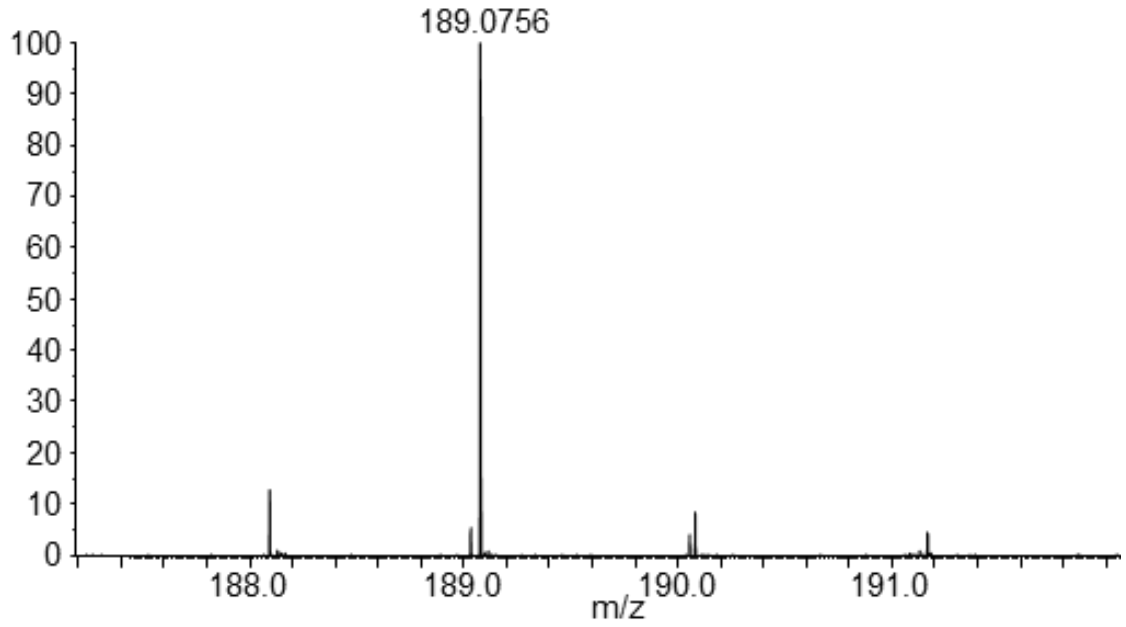
Spectrum	Observed m/z	Theoritical m/z	Error (ppm)
	189.07558	189.07574	0.88

Figure S7: **HRMS** of Compound **1,** Calculated $C_8H_{13}O_5$ $[M+H]^+$ 189.07574 | Found 189.0758

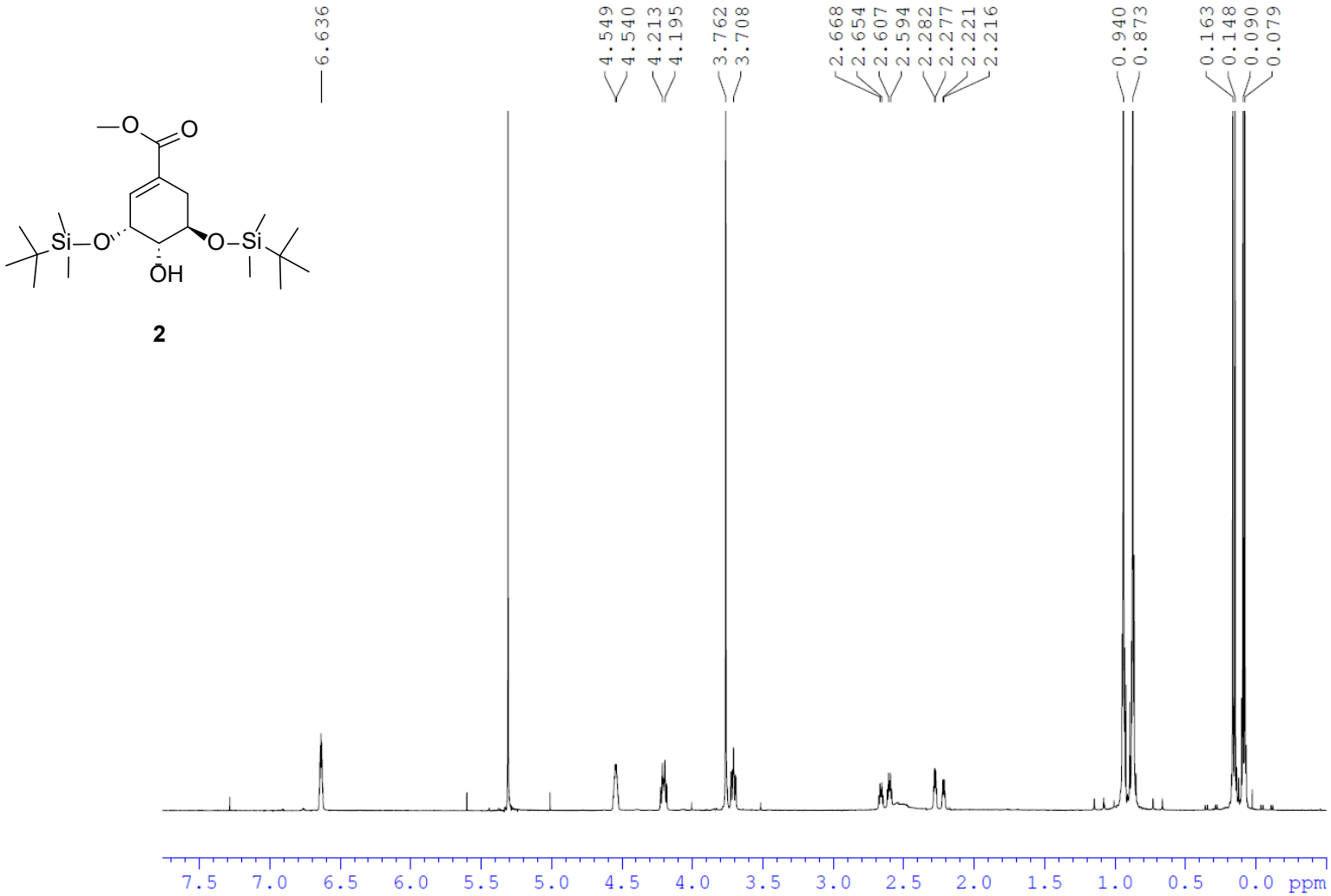
*Synthesis of methyl (3R,4S,5R)-3,5-bis((tert-butyldimethylsilyl)oxy)-4-hydroxycyclohex-1-ene-1-carboxylate **(2)***:

A 250 mL round bottom flask equipped with reflux condenser and stir bar was charged with dichloromethane (50 mL) and compound **1** (833 mg, 4.4 mmol, 1 equiv). The solution was stirred and heated under reflux for 2 hours to dissolve the shikimate ester. After being refluxed, the mixture was allowed to cool to room temperature then in an ice bath. The stirring was continued in ice. 2,6-lutidine (1.28 mL, 11 mmol, 2.5 equiv) was added to the solution and TBDMSOTf (2.3 mL, 10.1 mmol, 2.3 equiv) was gradually introduced to the resulting mixture, starting with 1 mL. After 2 hours, the remaining of the TBDMSOTf (1.3 mL) was added and the reaction mixture was stirred overnight at room temperature. TLC analysis showed a large proportion of desired double protected product (low R_f) and low level of triple protected product (high R_f). The crude mixture was purified by column chromatography over silica gel using dichloromethane as eluent to give 1.2 g (2.9 mmol, 65 %) of product 2 as a colourless oil. The ^{1}H and ^{13}C NMR spectra are consistent with the literature(*43*).

Figure S8: **^{1}H NMR (300 MHz, $CDCl_3$, 25 °C)** δ (ppm) 6.63 (dddd, J = 2.7, 2.5, 1.5, 1.4 Hz, 1H, H_2), 4.54 (m, 1H, H_3), 4.20 (dddd, J = 5.8, 4, 3.3, 0.5 Hz, 1H, H_5), 3.76 (s, 3H, COOMe), 3.70 (ddd, J = 5.4, 4, 1.3 Hz, 1H, H_4), 2.63 (ddtd, J = 18.3, 4.2, 2.7, 0.3 Hz, 1H, H_6), 2.24 (ddt, J= 18.4, 3.3, 1.5 Hz, 1H, $H_{6'}$), 0.94 (s, 9H, Me_3Csi), 0.87 (s, 9H, Me_3Csi), δ 0.16 (s, 3H, MeSi), 0.14 (s, 3H, MeSi), 0.09 (s, 3H, MeSi), 0.07 (s, 3H, MeSi).

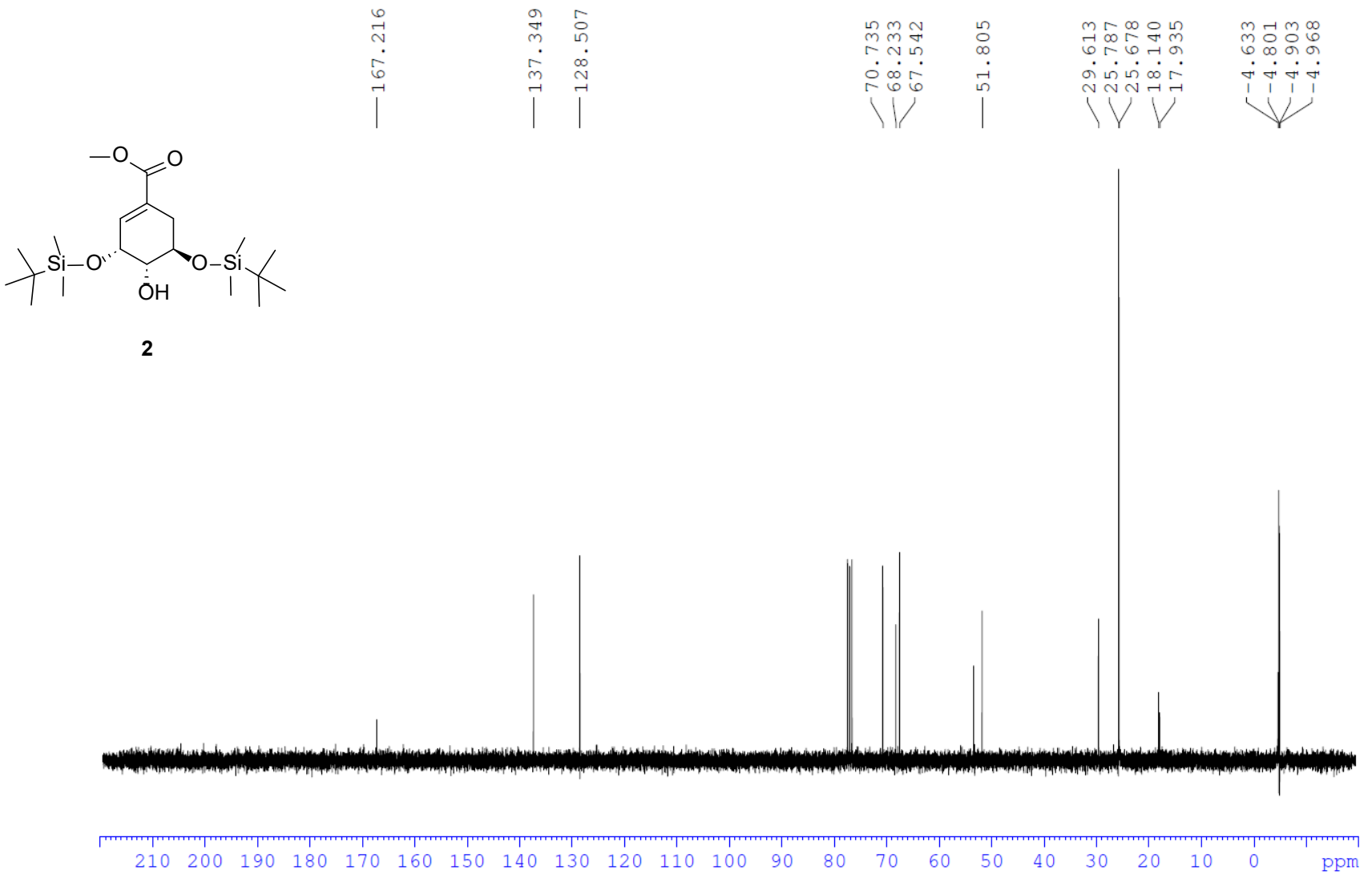
Figure S9: **^{13}C NMR (75 MHz, $CDCl_3$, 25 °C)** δ (ppm) 167.22, 137.35, 128.51, 70.74, 68.23, 67.54, 51.80, 29.61, 25.79, 25.68, 18.14, 17.93, -4.63, -4.80, -4.90, -4.96.

Figure S10: **HRMS** Calculated $C_{20}H_{39}O_5Si_2$ $[M-H]^+$ 415.23415 | Found 415.23462



2

Figure S8: Compound **2**, **^{1}H NMR (300 MHz, $CDCl_3$, 25 °C)** δ (ppm) 6.63 (dddd, J = 2.7, 2.5, 1.5, 1.4 Hz, 1H, H_2), 4.54 (m, 1H, H_3), 4.20 (dddd, J = 5.8, 4, 3.3, 0.5 Hz, 1H, H_5), 3.76 (s, 3H, COOMe), 3.70 (ddd, J = 5.4, 4, 1.3 Hz, 1H, H_4), 2.63 (ddtd, J = 18.3, 4.2, 2.7, 0.3 Hz, 1H, H_6), 2.24 (ddt, J= 18.4, 3.3, 1.5 Hz, 1H, $H_{6'}$), 0.94 (s, 9H, Me_3Csi), 0.87 (s, 9H, Me_3Csi), δ 0.16 (s, 3H, MeSi), 0.14 (s, 3H, MeSi), 0.09 (s, 3H, MeSi), 0.07 (s, 3H, MeSi).



2

Figure S9: Compound **2**, **^{13}C NMR (75 MHz, $CDCl_3$, 25 °C)** δ (ppm) 167.22, 137.35, 128.51, 70.74, 68.23, 67.54, 51.80, 29.61, 25.79, 25.68, 18.14, 17.93, -4.63, -4.80, -4.90, -4.96.

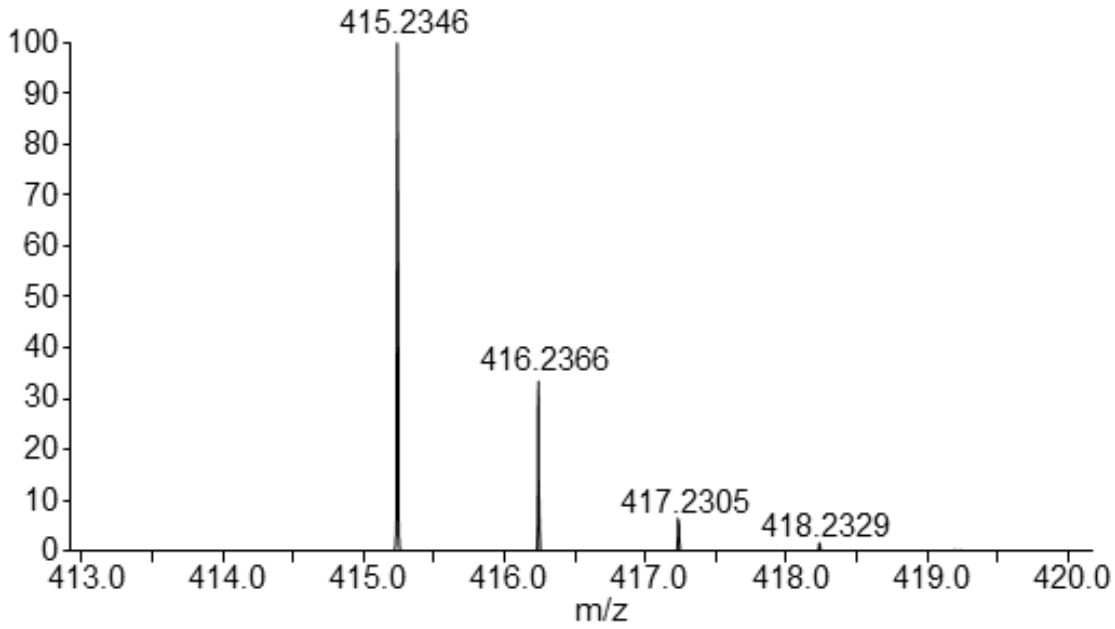
Spectrum	Observed m/z	Theoritical m/z	Error (ppm)
	415.23462	415.23415	1.1

Figure S10: **HRMS** of compound **2**, Calculated $C_{20}H_{39}O_5Si_2$ $[M-H]^+$ 415.23415 | Found 415.23462

*Synthesis of methyl (3R,5R)-3,5-bis((tert-butyldimethylsilyl)oxy)-4-oxocyclohex-1-ene-1-carboxylate **(3)***:

Dess-Martin Periodinane (14.5 mL, 4.35 mmol, 1.5 equiv) was added to **2** (1.2 g, 2.9 mmol, 1 equiv) in dichloromethane (40 mL). The solution was stirred at room temperature for two hours during which a white precipitate was gradually formed. Full product recovery was confirmed by TLC. The mixture was split into eppendorfs microtubes and centrifuged (1 min, 10000 rpm) at room temperature. The pellets containing the white precipitate were removed and the combined supernatants were purified via column chromatography over silica gel, using dichloromethane as eluent, to yield the final product as a colourless oil (542 mg, 1.3 mmol, 45 %).

Figure S11: **^{1}H NMR (300 MHz, $CDCl_3$, 25 °C)** δ (ppm) 6.74 (dt, J = 3.3, 1.5 Hz, 1H, H_2), 4.93 (ddd, J = 3, 1.8, 1 Hz, 1H, H_3), 4.4 (dd, J = 5.7, 5.1 Hz, H_5), 3.72 (s, 3H, COOMe), 2.9 (ddt, J = 18, 5.7, 1.2 Hz, 1H, H_6), 2.68 (ddd, J = 18, 5, 1.8 Hz, 1H, $H_{6'}$), 0.87 (s, 9H, Me_3Csi), 0.84 (s, 9H, Me_3Csi), 0.09 (s, 3H, MeSi), 0.07 (s, 3H, MeSi), 0.05 (s, 3H, MeSi), 0.03(s, 3H, MeSi).

Figure S12: **^{13}C NMR (75 MHz, $CDCl_3$, 25 °C)** δ (ppm) 206.50, 167.12, 139.95, 129.87, 73.46, 71.45, 53.02, 37.24, 26.57, 26.48, 19.21, 18.97, -3.87, -4.09, -4.33, -4.40.

Figure S13: **HRMS** Calculated $C_{20}H_{37}O_5Si_2$ $[M-H]^+$ 413.21849 | Found 413.21886

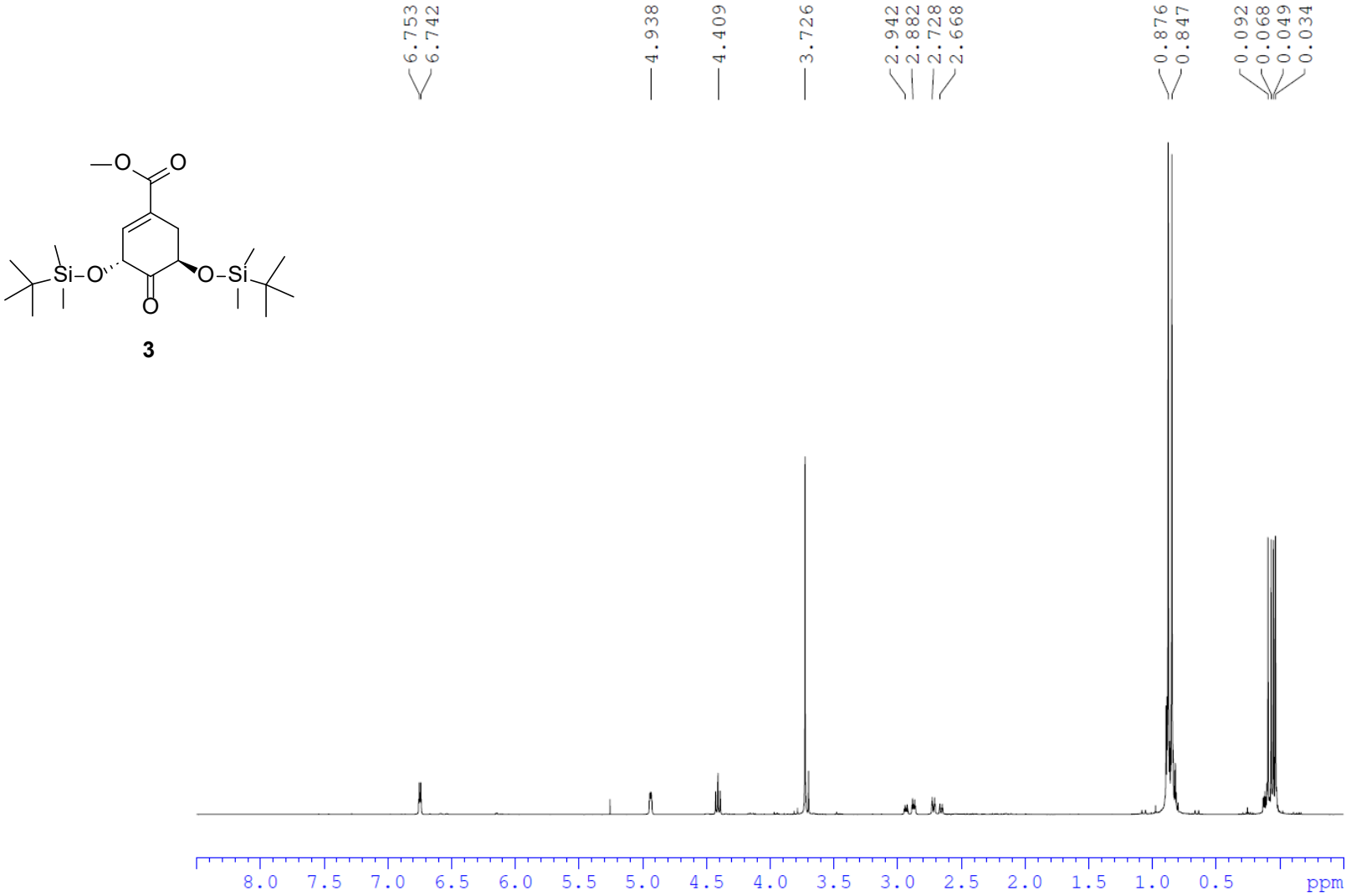


Figure S11: Compound **3**, **^{1}H NMR (300 MHz, $CDCl_3$, 25 °C)** δ (ppm) 6.74 (dt, J = 3.3, 1.5 Hz, 1H, H_2), 4.93 (ddd, J = 3, 1.8, 1 Hz, 1H, H_3), 4.4 (dd, J = 5.7, 5.1 Hz, H_5), 3.72 (s, 3H, COOMe), 2.9 (ddt, J = 18, 5.7, 1.2 Hz, 1H, H_6), 2.68 (ddd, J = 18, 5, 1.8 Hz, 1H, $H_{6'}$), 0.87 (s, 9H, Me_3Csi), 0.84 (s, 9H, Me_3Csi), 0.09 (s, 3H, MeSi), 0.07 (s, 3H, MeSi), 0.05 (s, 3H, MeSi), 0.03(s, 3H, MeSi).

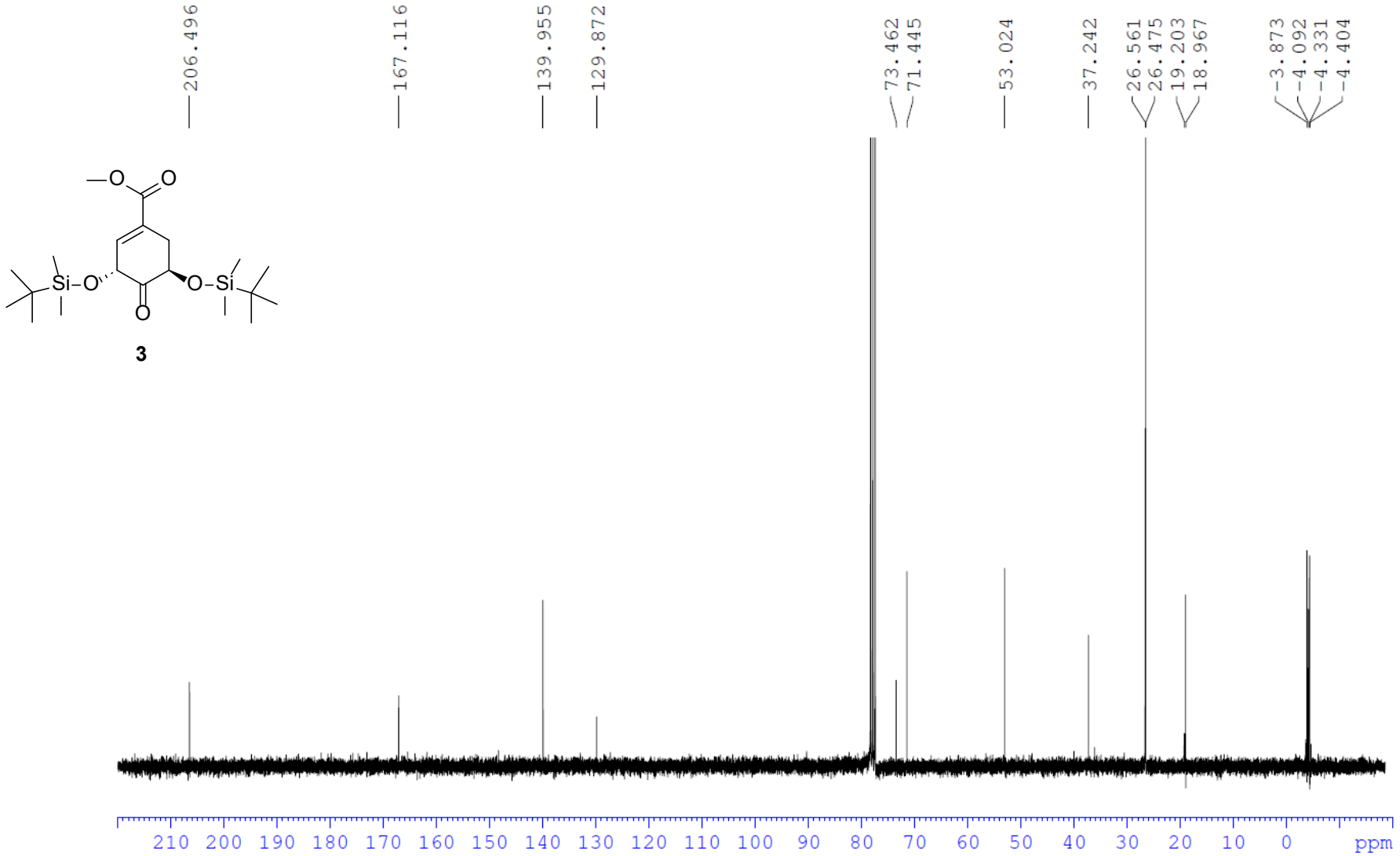


Figure S12: Compound **3**, **^{13}C NMR (75 MHz, $CDCl_3$, 25 °C)** δ (ppm) 206.50, 167.12, 139.95, 129.87, 73.46, 71.45, 53.02, 37.24, 26.57, 26.48, 19.21, 18.97, -3.87, -4.09, -4.33, -4.40.

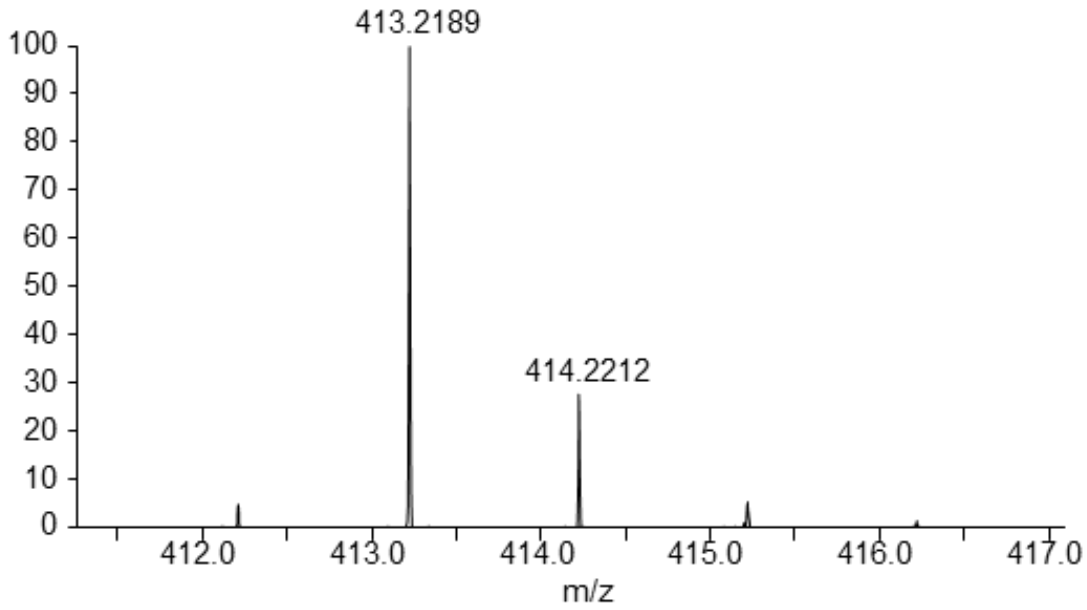
Spectrum	Observed m/z	Theoritical m/z	Error (ppm)
	413.21886	413.21849	0.9

Figure S13: **HRMS** of Compound **3,** Calculated $C_{20}H_{37}O_5Si_2$ $[M-H]^+$ 413.21849 | Found 413.21886

*Synthesis of methyl (3R,5R)-3,5-bis((tert-butyldimethylsilyl)oxy)-4-[^{18}O]oxocyclohex-1-ene-1-carboxylate **(4)***:

The previously obtained ketone **3** (100mg, 0.24 mmol, 1 equiv) was suspended in dry THF (0.8 mL). Then, $H_2{}^{18}O$ (0,4 mL) and acetic acid-$^{18}O_2$ (5 µL, 0.09 mmol), as a catalyst, were added and the resulting mixture was stirred for 3 days at room temperature. The solvent was removed by rotary evaporation. To avoid $^{18}O/^{16}O$ exchange during purification and to prevent the loss of oxygen-18 labelling, the crude compound 4 was used without further purification in the following steps.

*Synthesis of C_4[^{18}O] shikimate, (3R,4S,5R)-3,4[^{18}O],5-trihydroxy-4-cyclohex-1-ene-1-carboxylic acid **(5)***:

To a solution of ^{18}O labelled ketone **4** (100mg, 0.24 mmol, 1 equiv) from the previous step in THF (900 µL) and H_2O (100 µL), was introduced $NaBH_4$ (480 µL, 0.24 mmol, 1equiv). The reaction mixture was stirred at room temperature for 1 hour. The solvent was evaporated and the residue was suspended in H_20 (2 mL) and THF (2 mL). To this solution was added APTS (10 mg, 0.05 mmol) as a catalyst. The resulting mixture was stirred at 70°C for 4 days. The crude was purified by HPLC (the expected product is eluted at 13.6 min) to obtain the final product as a white solid (4.25 mg, 0.024 mmol). The yield for the last two steps is 10 %.

The ^{13}C NMR spectrum of compound **5** shows two chemical shifts (71.30 and 71.32 ppm) of the carbon at position C_4, corresponding to C_4[^{18}O]-shikimate and C_4[^{16}O]-shikimate respectively (superimposition Fig. S13 and S14). It has been reported that the chemical shift of a carbon atom carrying an oxygen isotope can be slightly modified (*46*, *47*).

Figure S14: **^{1}H NMR** (**300 MHz, CD_3OD, 25 °C**) δ (ppm) 6.81 (dtd, J = 3.8, 1.9, 0.6 Hz, 1H, H_2), 4.38 (m, 1H, H_3), 4.0 (dddd, J = 7.4, 5.6, 4.9, 0.5 Hz, 1H, H_5), 3.70 (dd, J = 7.2, 4.2 Hz, 1H, H_4), 2.71 (ddtd, J = 18.2, 4.9, 1.8, 0.3 Hz, 1H, H_6), 2.21 (ddt, J = 18.1, 5.5, 1.7 Hz, 1H, $H_{6'}$).

Figure S15: **^{13}C NMR (75 MHz, CD_3OD, 25 °C**) δ (ppm) 168.66, 137.46, 129.32, 71.32, 71.30, 67.02, 65.93, 30.23.

Figure S16: Overlay of **^{1}H NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and of Compound **5** (blue trace)

Figure S17: Overlay of **^{13}C NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and Compound **5** (blue trace)

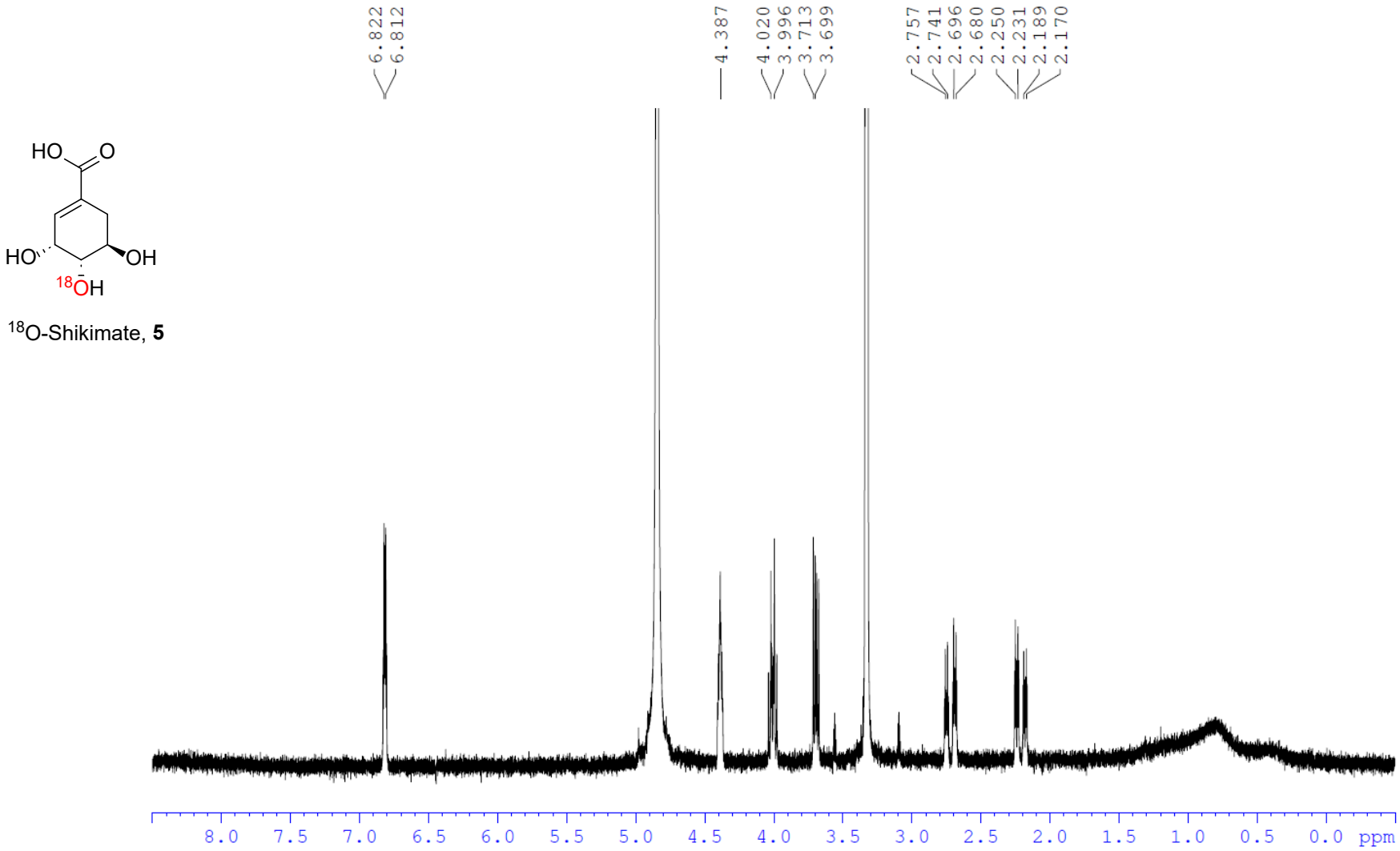
Figure S18: Zoom into the 70-72.5 ppm chemical shifts of the overlay of **^{13}C NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and Compound **5** (blue trace)

Figure S19: **HRMS** of Compound 5, Calculated $C_7H_9O_4{}^{18}O$ $[M-H]^+$ 175.04870 | Found 175.04880
High-resolution mass spectrometry analysis of the synthesised $C_4[^{18}O]$-shikimate determined an Oxygen 18 labelling rate of approximately 63 ± 3%.

Figure S20: **HRMS** of commercial Shikimate, Calculated $C_7H_9O_5$ $[M-H]^+$ 173.04445 | Found 173.04449

Figure S21: **MS/MS** spectrum of Compound 5

Figure S22: Overlay of **MS/MS** spectra of Compound 5 and of commercial Shikimate



^{18}O-Shikimate, **5**

Figure S14: Compound **5**, **^{1}H NMR (300 MHz, CD_3OD, 25 °C)** δ (ppm) 6.81 (dtd, J = 3.8, 1.9, 0.6 Hz, 1H, H_2), 4.38 (m, 1H, H_3), 4.0 (dddd, J = 7.4, 5.6, 4.9, 0.5 Hz, 1H, H_5), 3.70 (dd, J = 7.2, 4.2 Hz, 1H, H_4), 2.71 (ddtd, J = 18.2, 4.9, 1.8, 0.3 Hz, 1H, H_6), 2.21 (ddt, J = 18.1, 5.5, 1.7 Hz, 1H, $H_{6'}$).

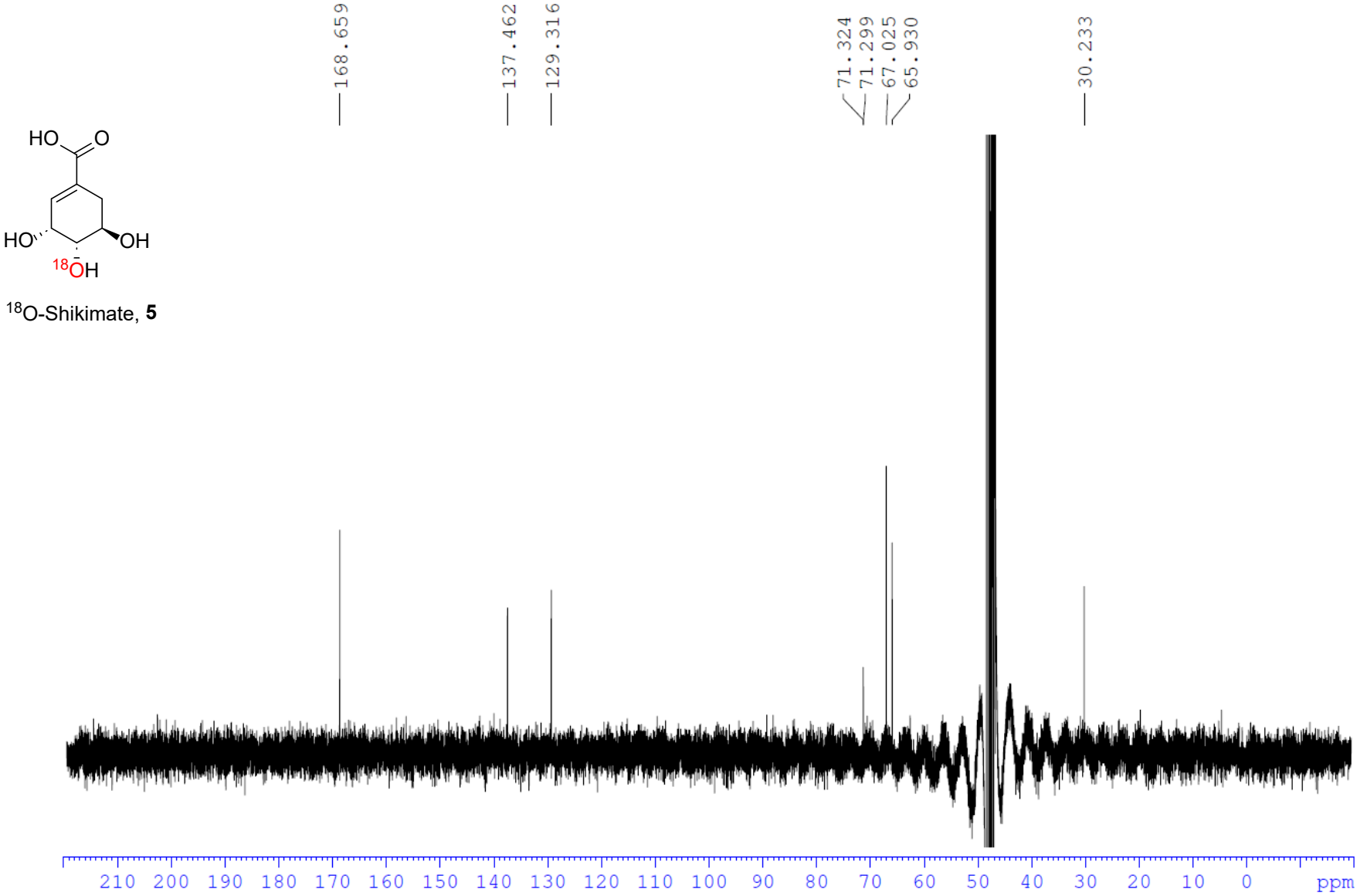


Figure S15: Compound **5**, **^{13}C NMR (75 MHz, CD_3OD, 25 °C)** δ (ppm) 168.66, 137.46, 129.32, 71.32, 71.30, 67.02, 65.93, 30.23.

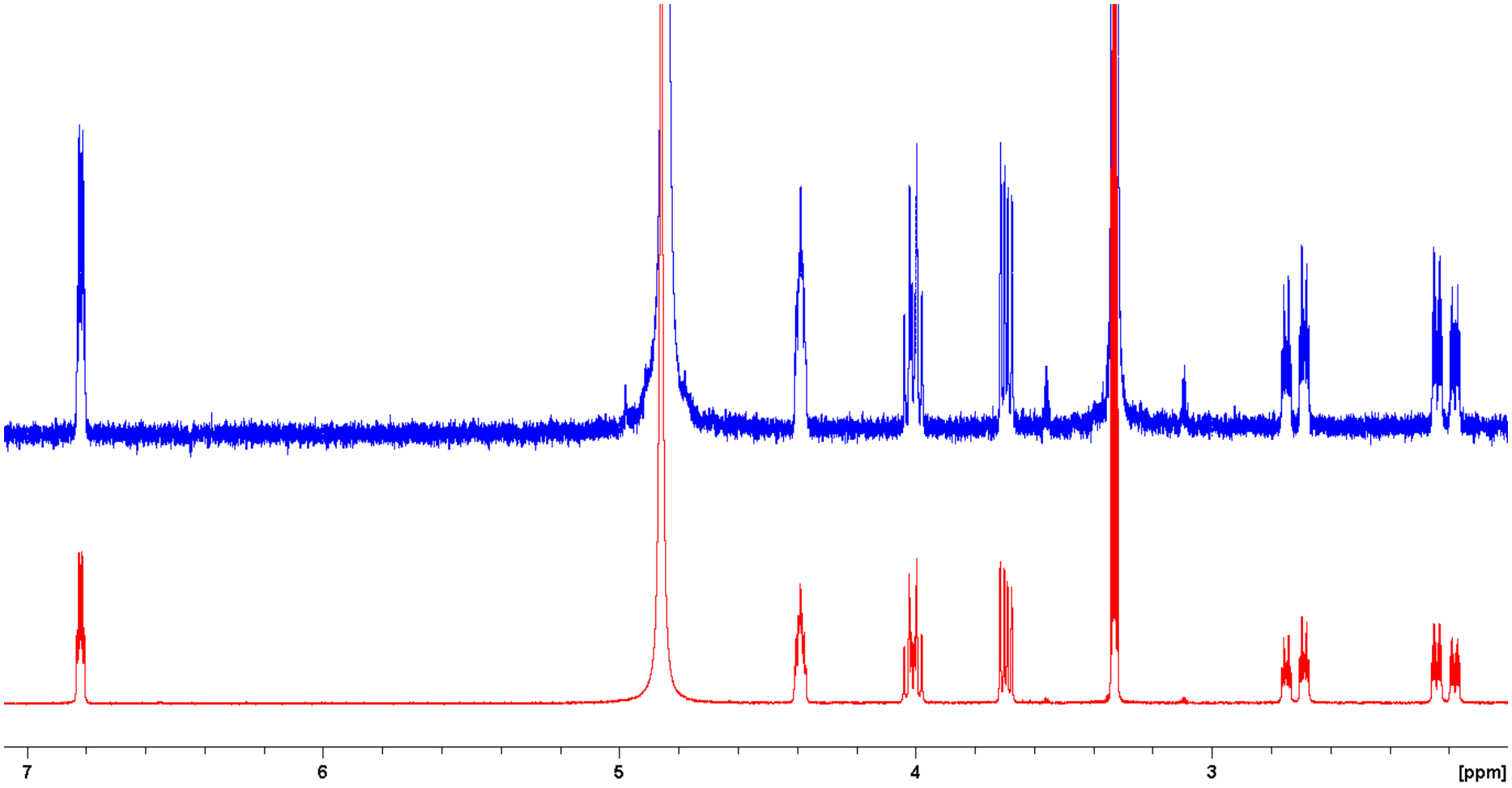


Figure S16: Overlay of **^{1}H NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and of Compound **5** (blue trace)

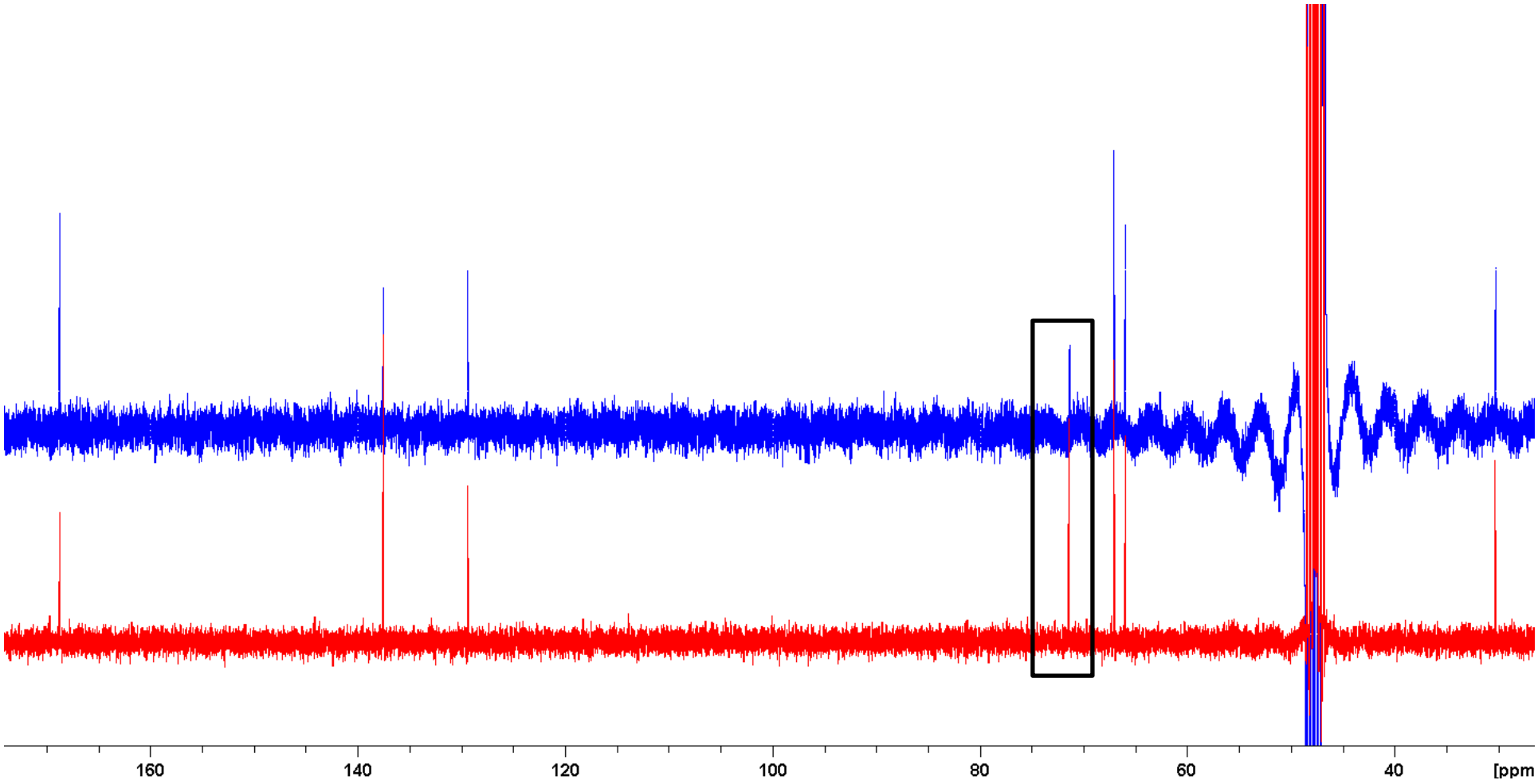


Figure S17: Overlay of **^{13}C NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and Compound **5** (blue trace). The black rectangle corresponds to the 70-72.5 ppm region zoomed in Figure S14.

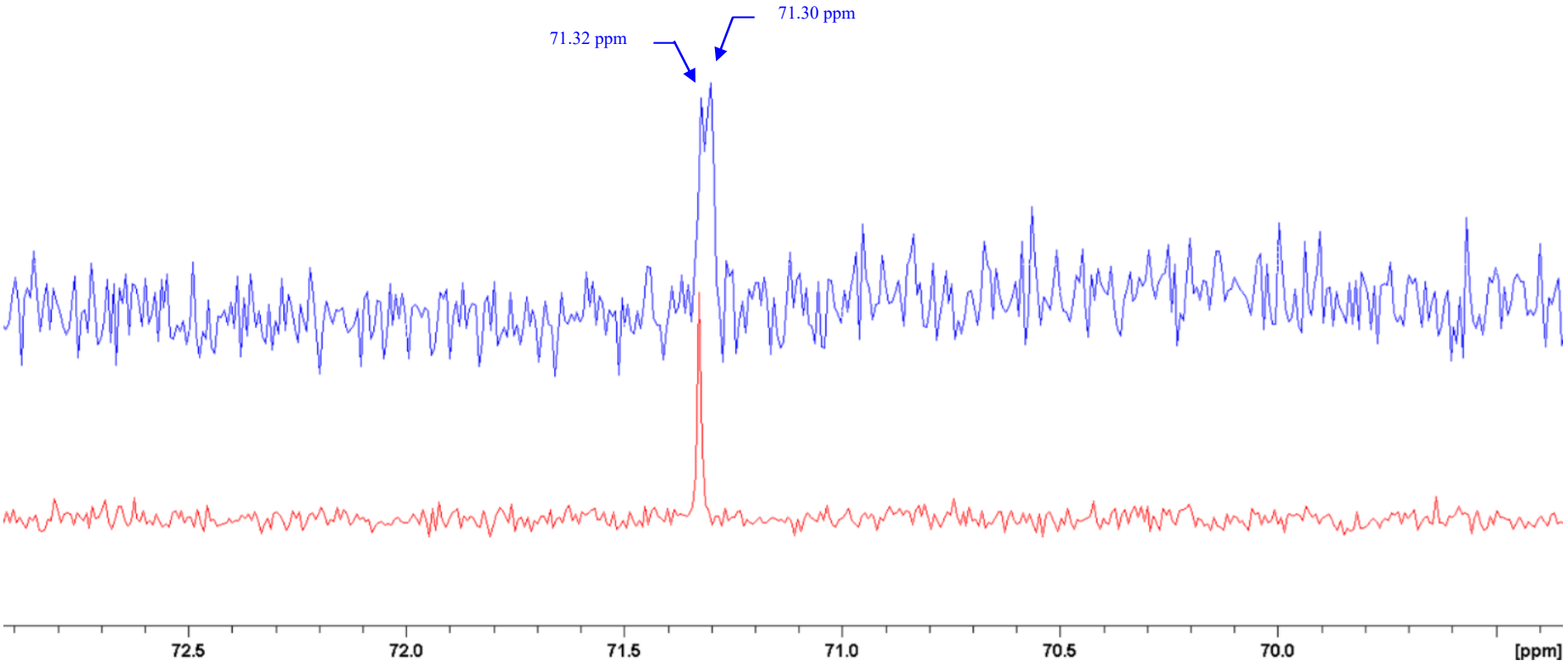


Figure S18: Zoom into the 70-72.5 ppm chemical shifts of the overlay of **^{13}C NMR** spectra (300 MHz, CD_3OD) of commercial Shikimate (red trace) and Compound **5** (blue trace).

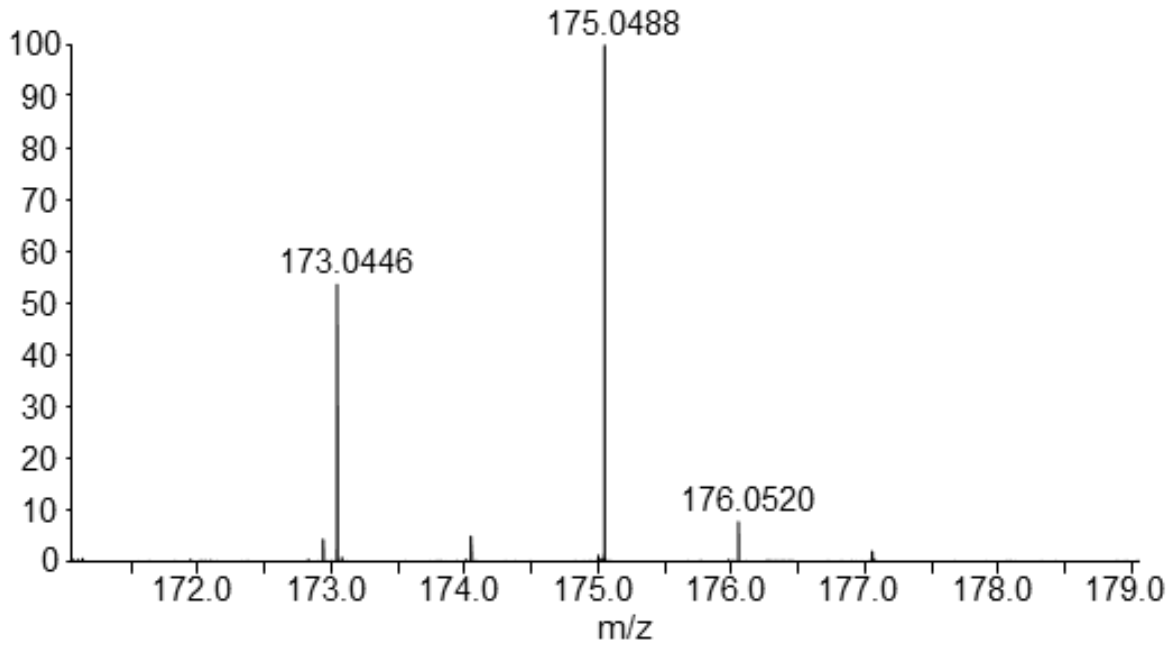
Spectrum	Observed m/z	Theoritical m/z	Error (ppm)
	175.04880	175.04870	0.57

Figure S19: **HRMS** of Compound **5**, Calculated $C_7H_9O_4{}^{18}O$ $[M-H]^+$ 175.04870 | Found 175.04880

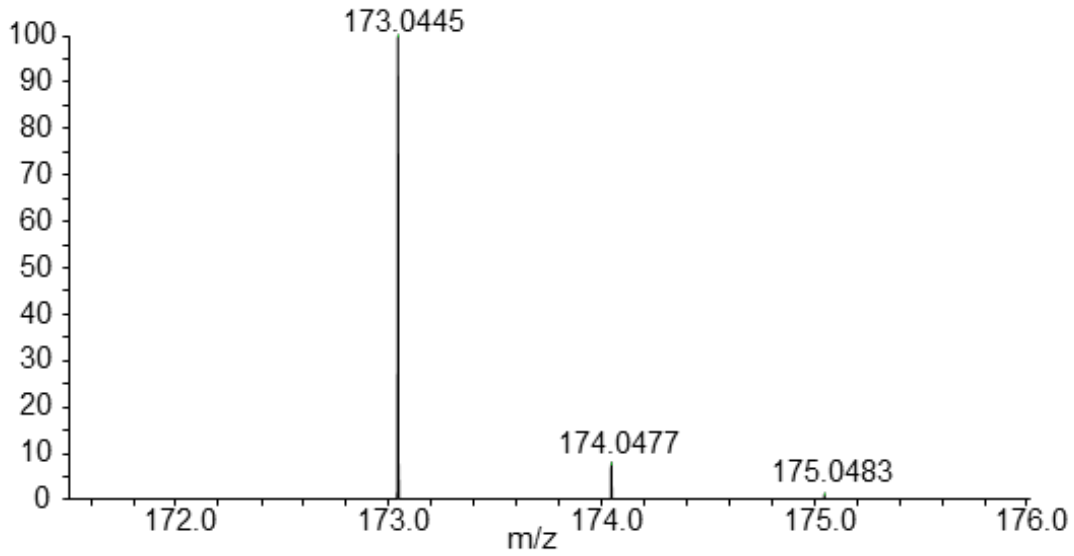
Spectrum	Observed m/z	Theoritical m/z	Error (ppm)
	173.04449	173.04445	0.02

Figure S20: **HRMS** of commercial Shikimate, Calculated $C_7H_9O_5$ $[M-H]^+$ 173.04445 | Found 173.04449

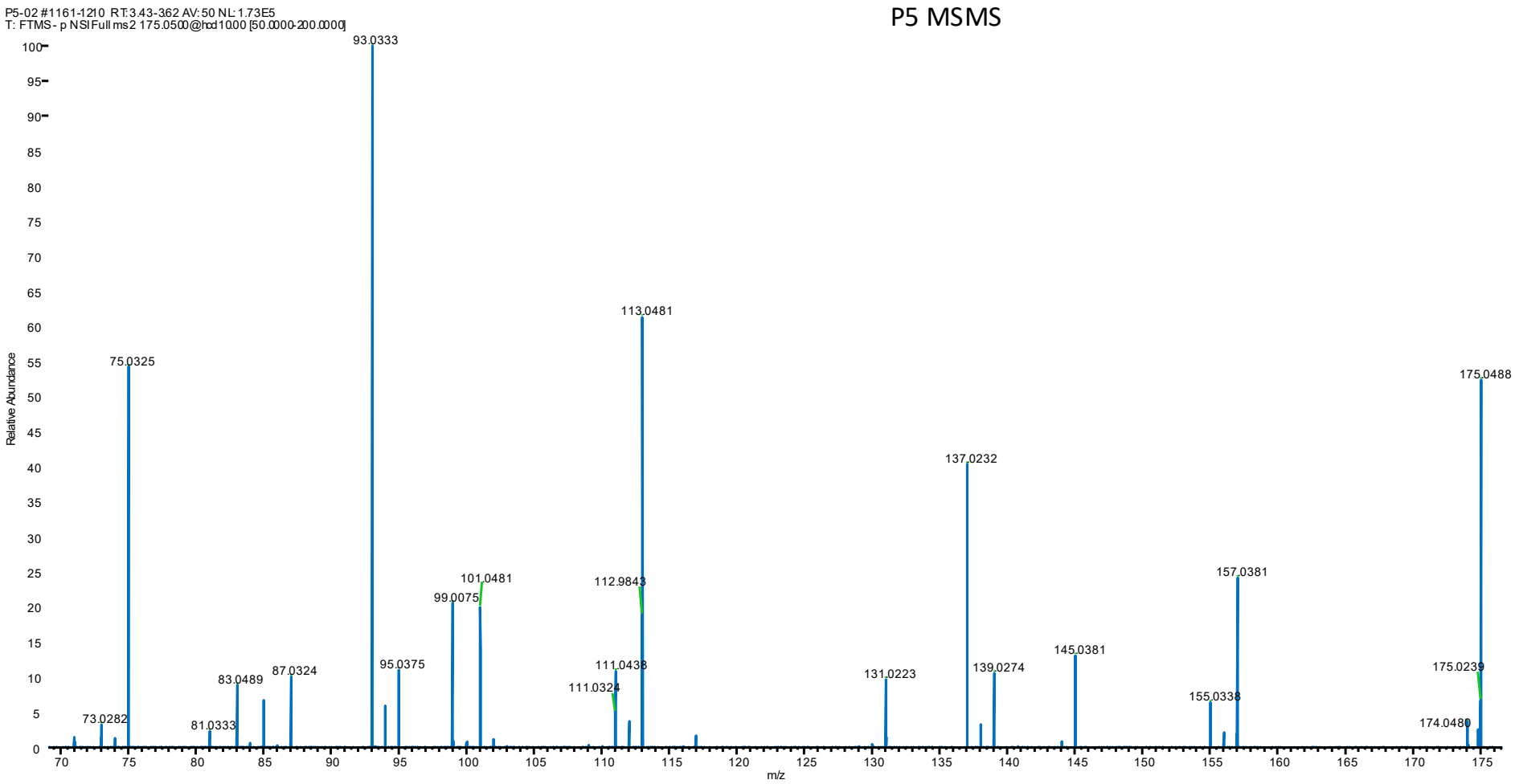


Figure S21: MS/MS of compound **5**

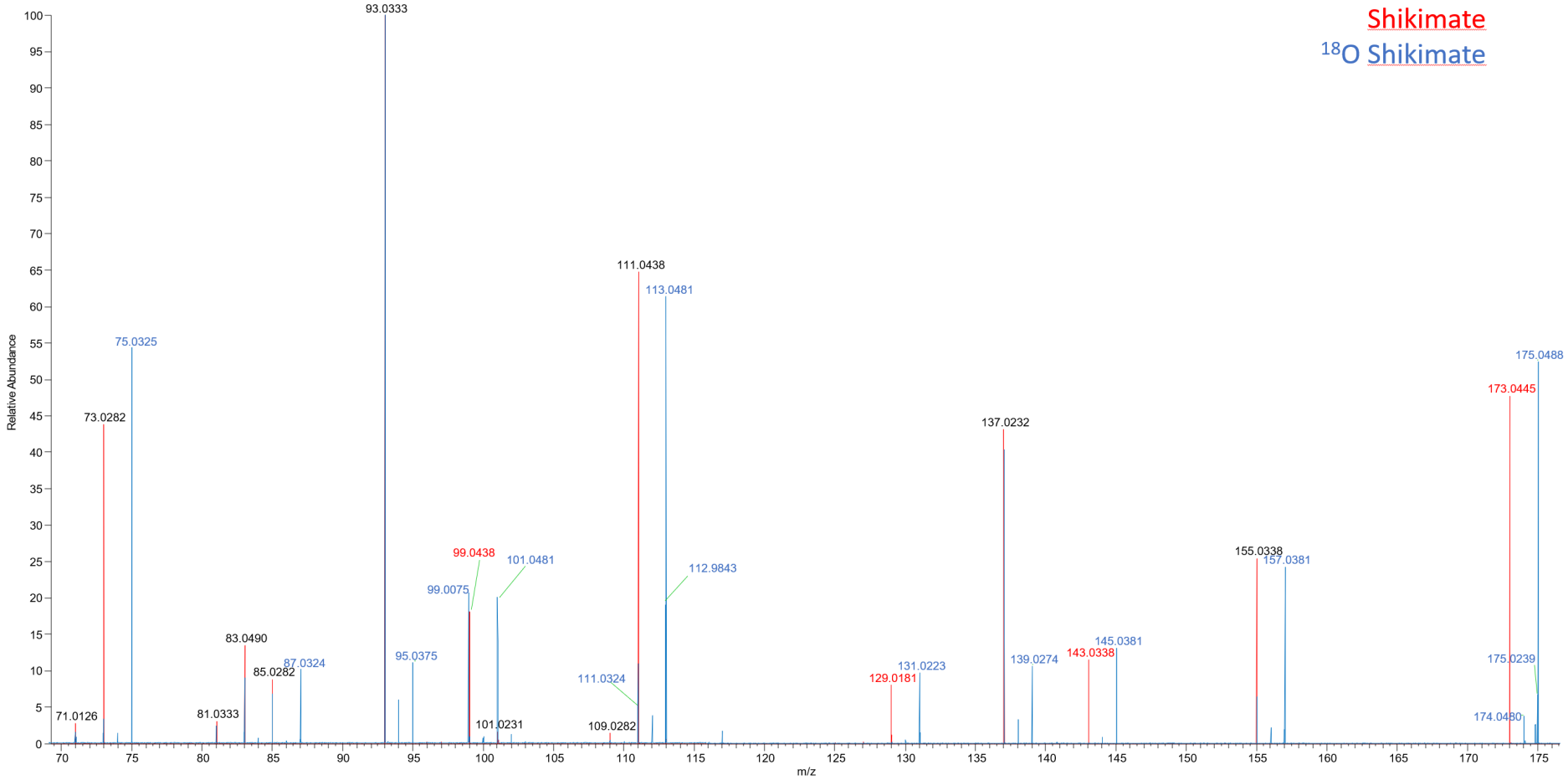


Figure S22: Overlay of commercial Shikimate (red) and C4[^{18}O]-shikimate **5** (blue) HCD 10 spectra. Masses are labelled in red, blue or black depending if the mass is present in Shikimate, ^{18}O Shikimate or both spectra respectively.

Supplementary references

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